

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Math Hub | Mathematics Reviewer</title>

<style>

* {
  box-sizing: border-box;
}

:root {
  --bg: #f5f3fb;
  --card: #ffffff;
  --text: #29233d;
  --muted: #746d87;
  --border: #e5def2;
  --primary: #7654d9;
  --primary-dark: #5d3db5;
  --soft: #eee9fc;
  --header: #302052;
  --success: #16804b;
  --danger: #c0392b;
}

body.dark {
  --bg: #15121f;
  --card: #211c30;
  --text: #f4efff;
  --muted: #b8afcb;
  --border: #403653;
  --primary: #a78bfa;
  --primary-dark: #8b6be8;
  --soft: #302746;
  --header: #211735;
}

body {
  margin: 0;
  font-family: Arial, Helvetica, sans-serif;
  background: var(--bg);
  color: var(--text);
  transition: background 0.3s, color 0.3s;
}

button,
input {
  font: inherit;
}

button {
  cursor: pointer;
}
```

```
button:focus-visible,
input:focus-visible {
  outline: 3px solid var(--primary);
  outline-offset: 3px;
}

/* HEADER */

header {
  background: linear-gradient(135deg, #302052, #6543a5);
  color: white;
  padding: 28px 20px 34px;
}

.header-inner {
  max-width: 1050px;
  margin: auto;
}

.topbar {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 15px;
}

.logo {
  display: flex;
  align-items: center;
  gap: 12px;
}

.logo-icon {
  background: #9b7de8;
  width: 45px;
  height: 45px;
  border-radius: 13px;
  display: grid;
  place-items: center;
  font-size: 25px;
  font-weight: bold;
}

.logo h2 {
  font-size: 18px;
  margin: 0;
}

.logo p {
  margin: 4px 0 0;
  color: #d8cef3;
  font-size: 12px;
}

.theme-btn {
  border: 1px solid #ffffff55;
```

```

    background: #ffffff18;
    color: white;
    border-radius: 20px;
    padding: 9px 14px;
    cursor: pointer;
}

.hero {
    margin-top: 35px;
}

.eyebrow {
    font-size: 11px;
    font-weight: bold;
    letter-spacing: 2px;
    color: #d4c6f4;
}

.hero h1 {
    font-size: clamp(28px, 5vw, 46px);
    line-height: 1.15;
    margin: 12px 0;
}

.hero p {
    color: #ded5f4;
    line-height: 1.6;
    max-width: 650px;
}

.badges {
    display: flex;
    gap: 8px;
    flex-wrap: wrap;
    margin-top: 20px;
}

.badge {
    background: #ffffff20;
    border: 1px solid #ffffff20;
    padding: 8px 13px;
    border-radius: 20px;
    font-size: 12px;
}

/* MAIN */

main {
    max-width: 1050px;
    margin: auto;
    padding: 25px 20px 40px;
}

.search-box {
    display: flex;
    align-items: center;
    gap: 10px;

```

```
    background: var(--card);
    border: 1px solid var(--border);
    border-radius: 14px;
    padding: 14px 16px;
    margin-bottom: 25px;
}
```

```
.search-box input {
    border: none;
    outline: none;
    background: transparent;
    color: var(--text);
    width: 100%;
    min-width: 0;
}
```

```
.search-box input::placeholder {
    color: var(--muted);
}
```

```
.section-heading {
    display: flex;
    align-items: center;
    justify-content: space-between;
    gap: 10px;
    margin-bottom: 15px;
}
```

```
.section-heading h2 {
    margin: 0;
    font-size: 24px;
}
```

```
.section-heading p {
    color: var(--muted);
    font-size: 13px;
    margin: 6px 0 0;
}
```

```
.topic-count {
    background: var(--soft);
    color: var(--primary-dark);
    padding: 7px 12px;
    border-radius: 20px;
    font-size: 12px;
    font-weight: bold;
    white-space: nowrap;
}
```

```
/* FILTERS */
```

```
.filters {
    display: flex;
    gap: 8px;
    flex-wrap: wrap;
    margin-bottom: 20px;
}
```

```
.filter-btn {  
  border: 1px solid var(--border);  
  background: var(--card);  
  color: var(--text);  
  border-radius: 22px;  
  padding: 9px 15px;  
  font-size: 13px;  
}
```

```
.filter-btn.active {  
  background: var(--primary);  
  color: white;  
  border-color: var(--primary);  
}
```

```
/* LESSONS */
```

```
.lesson {  
  background: var(--card);  
  border: 1px solid var(--border);  
  border-radius: 15px;  
  margin-bottom: 12px;  
  overflow: hidden;  
}
```

```
.lesson.hidden {  
  display: none;  
}
```

```
.lesson-header {  
  width: 100%;  
  display: flex;  
  align-items: center;  
  gap: 13px;  
  border: none;  
  background: transparent;  
  color: var(--text);  
  padding: 17px;  
  text-align: left;  
}
```

```
.lesson-header:hover {  
  background: var(--soft);  
}
```

```
.lesson-number {  
  flex-shrink: 0;  
  width: 45px;  
  height: 45px;  
  border-radius: 13px;  
  background: var(--soft);  
  color: var(--primary-dark);  
  display: grid;  
  place-items: center;  
  font-weight: bold;  
  font-size: 14px;
```

```
}

.lesson-info {
  flex: 1;
  min-width: 0;
}

.lesson-name {
  font-size: 16px;
  font-weight: bold;
  line-height: 1.4;
}

.lesson-description {
  font-size: 12px;
  color: var(--muted);
  margin-top: 4px;
}

.arrow {
  font-size: 23px;
  color: var(--muted);
  transition: transform 0.2s;
}

.lesson.open .arrow {
  transform: rotate(180deg);
}

.lesson-content {
  display: none;
  padding: 0 20px 22px;
  border-top: 1px solid var(--border);
}

.lesson.open .lesson-content {
  display: block;
}

.lesson-content h3 {
  color: var(--primary-dark);
  font-size: 19px;
  margin: 22px 0 10px;
}

.lesson-content p {
  line-height: 1.7;
  font-size: 14px;
  margin: 8px 0;
}

.lesson-content ul {
  line-height: 1.8;
  font-size: 14px;
}

.formula {
```

```

background: var(--soft);
border: 1px solid var(--border);
border-radius: 10px;
padding: 14px;
margin: 12px 0;
font-size: 18px;
font-family: Georgia, serif;
overflow-wrap: anywhere;
}

.note {
background: var(--soft);
border-left: 4px solid var(--primary);
padding: 12px 14px;
border-radius: 0 8px 8px 0;
font-size: 13px;
line-height: 1.7;
margin: 13px 0;
}

.example {
background: var(--card);
border: 1px solid var(--border);
padding: 12px 14px;
border-radius: 10px;
margin: 10px 0;
line-height: 1.7;
font-size: 14px;
}

/* TABLE */

table {
border-collapse: collapse;
width: 100%;
margin: 12px 0;
font-size: 13px;
}

th,
td {
border: 1px solid var(--border);
padding: 10px;
text-align: left;
line-height: 1.5;
}

th {
background: var(--soft);
font-weight: bold;
}

/* VENN DIAGRAM */

.venn-wrapper {
background: var(--soft);
border: 1px solid var(--border);

```

```

    border-radius: 12px;
    padding: 10px;
    margin: 15px 0;
}

.venn-wrapper svg {
    width: 100%;
    max-width: 480px;
    display: block;
    margin: auto;
}

/* QUIZ */

.quiz-box {
    margin-top: 30px;
    background: var(--card);
    border: 1px solid var(--border);
    border-radius: 18px;
    padding: 22px;
}

.quiz-box h2 {
    margin-top: 0;
}

.quiz-subtitle {
    color: var(--muted);
    font-size: 13px;
}

.progress-track {
    height: 8px;
    background: var(--border);
    border-radius: 20px;
    overflow: hidden;
    margin: 18px 0;
}

.progress-bar {
    height: 100%;
    width: 0%;
    background: var(--primary);
    transition: width 0.2s;
}

.quiz-question {
    font-size: 16px;
    font-weight: bold;
    line-height: 1.6;
    margin: 15px 0;
}

.option {
    display: block;
    width: 100%;
    padding: 12px 14px;

```

```

    margin: 8px 0;
    text-align: left;
    border: 1px solid var(--border);
    border-radius: 10px;
    background: var(--card);
    color: var(--text);
    font-size: 14px;
}

.option:hover:not(:disabled) {
    background: var(--soft);
}

.option.correct {
    background: #dcfce7;
    color: #14532d;
    border-color: #22c55e;
}

.option.wrong {
    background: #fee2e2;
    color: #7f1d1d;
    border-color: #ef4444;
}

.quiz-feedback {
    font-size: 13px;
    line-height: 1.6;
    margin: 12px 0;
}

.quiz-btn {
    background: var(--primary);
    color: white;
    border: none;
    padding: 11px 18px;
    border-radius: 9px;
    font-weight: bold;
}

.quiz-btn:hover {
    background: var(--primary-dark);
}

.hidden {
    display: none !important;
}

.no-results {
    text-align: center;
    padding: 30px;
    color: var(--muted);
}

/* FOOTER */

footer {

```

```
background: var(--header);
color: white;
text-align: center;
padding: 30px 20px;
margin-top: 10px;
}
```

```
footer p {
  color: #cfc4e9;
  font-size: 13px;
  margin: 7px 0;
}
```

```
@media (max-width: 600px) {
```

```
  header {
    padding: 22px 15px 30px;
  }
```

```
  main {
    padding: 20px 12px 30px;
  }
```

```
  .hero h1 {
    font-size: 29px;
  }
```

```
  .lesson-header {
    padding: 14px;
  }
```

```
  .lesson-number {
    width: 39px;
    height: 39px;
    font-size: 12px;
  }
```

```
  .lesson-name {
    font-size: 14px;
  }
```

```
  .lesson-content {
    padding: 0 15px 20px;
  }
```

```
  .section-heading h2 {
    font-size: 21px;
  }
```

```
  .topic-count {
    font-size: 11px;
  }
```

```
  th,
  td {
    padding: 8px;
    font-size: 12px;
  }
```

```

    }

    .topbar {
      align-items: flex-start;
    }
  }

</style>
</head>

<body>

<!-- HEADER -->

<header>

<div class="header-inner">

  <div class="topbar">

    <div class="logo">

      <div class="logo-icon"> $\Sigma$ </div>

      <div>
        <h2>MATH HUB</h2>
        <p>Student Reviewer</p>
      </div>

    </div>

    <button class="theme-btn" id="themeBtn">
      ☾ Dark Mode
    </button>

  </div>

<div class="hero">

  <div class="eyebrow">
    MATHEMATICS IN THE MODERN WORLD
  </div>

  <h1>
    Learn. Practice.<br>
    Understand Mathematics.
  </h1>

  <p>
    Your personal mathematics reviewer.
    Explore mathematical language, expressions,
    sets, Venn diagrams, and reasoning.
  </p>

  <div class="badges">

    <span class="badge">📖 12 Lessons</span>

```

```
<span class="badge">✎ Notes & Examples</span>

<span class="badge">📝 Practice Quiz</span>

</div>

</div>

</div>

</header>

<!-- MAIN -->

<main>

  <div class="search-box">

    <span>🔍</span>

    <input
      type="search"
      id="searchInput"
      placeholder="Search lessons, symbols, sets..."
    >

  </div>

  <div class="section-heading">

    <div>

      <h2>Your Lessons</h2>

      <p>
        Based on your handwritten notes
      </p>

    </div>

    <span class="topic-count" id="topicCount">
      12 Topics
    </span>

  </div>

  <div class="filters">

    <button class="filter-btn active" data-filter="all">
      All Topics
    </button>
```

```
<button class="filter-btn" data-filter="language">
  Symbols
</button>
```

```
<button class="filter-btn" data-filter="expression">
  Expressions
</button>
```

```
<button class="filter-btn" data-filter="sets">
  Sets
</button>
```

```
<button class="filter-btn" data-filter="reasoning">
  Reasoning
</button>
```

```
</div>
```

```
<!-- LESSON LIST -->
```

```
<div id="lessonList">
```

```
<!-- LESSON 1 -->
```

```
<section class="lesson open"
data-category="language"
data-search="mathematical language symbols addition subtraction multiplication division equal
greater less variables square root sigma">
```

```
<button class="lesson-header">
```

```
  <div class="lesson-number">01</div>
```

```
  <div class="lesson-info">
```

```
    <div class="lesson-name">
      Mathematical Language and Symbols
    </div>
```

```
    <div class="lesson-description">
      Common mathematical symbols
    </div>
```

```
  </div>
```

```
  <span class="arrow">v </span>
```

```
</button>
```

```
<div class="lesson-content">
```

```
<h3>What is Mathematical Language?</h3>
```

```
<p>
```

Mathematical language is a way of expressing ideas using numbers, symbols, variables, and mathematical terms.

Common Mathematical Symbols

||
||
||

Symbol

Meaning

+	Addition
---	----------

-	Subtraction
---	-------------

x or ·	Multiplication
--------	----------------

÷ or /	Division
--------	----------

=	Equal to
---	----------

≠	Not equal to
---	--------------

>	Greater than
---	--------------

<	Less than
---	-----------

≥	Greater than or equal to
---	--------------------------

≤	Less than or equal to
---	-----------------------

$\sqrt{\quad}$	Square root
----------------	-------------

$\sqrt[3]{\quad}$	Cube root
-------------------	-----------

Σ	Sigma / Summation
----------	-------------------

x, y, z	Variables
---------	-----------

Example

$$5 + 3 = 8$$

The symbol + means addition, and = means equal to.

<!-- LESSON 2 -->

02

```
<div class="lesson-name">
  Mathematical Expressions
</div>
```

```
<div class="lesson-description">
  Verbal phrases into mathematical symbols
</div>
```

```
</div>
```

```
<span class="arrow">v </span>
```

```
</button>
```

```
<div class="lesson-content">
```

```
<h3>Definition</h3>
```

```
<p>
A mathematical expression is a combination of numbers,
variables, and mathematical operations.
</p>
```

```
<div class="formula">
  x + 10
</div>
```

```
<p>
Here, x is the variable and 10 is a constant.
</p>
```

```
<h3>Verbal Phrase to Mathematical Expression</h3>
```

```
<table>
```

```
<tr>
<th>Verbal Phrase</th>
<th>Expression</th>
</tr>
```

```
<tr>
<td>The sum of a number and 10</td>
<td>x + 10</td>
</tr>
```

```
<tr>
<td>The difference of a number and 5</td>
<td>x - 5</td>
</tr>
```

```
<tr>
<td>The product of 10 and a number</td>
<td>10x</td>
</tr>
```

```
<tr>
```

<td>The quotient of a number and 50</td>
<td> $x / 50$ </td>
</tr>

<tr>
<td>One-fifth of a number</td>
<td> $x / 5$ </td>
</tr>

<tr>
<td>Three-fourths of 500</td>
<td> $(3/4)(500) = 375$ </td>
</tr>

<tr>
<td>The sum of two numbers less 11</td>
<td> $x + y - 11$ </td>
</tr>

<tr>
<td>Double the quantity of sales less taxes</td>
<td> $2x - y$ </td>
</tr>

<tr>
<td>The difference of gross sales less taxes</td>
<td> $x - y$ </td>
</tr>

<tr>
<td>A number subtracted from 8</td>
<td> $8 - x$ </td>
</tr>

</table>

<div class="note">

Remember:

"A number subtracted from 8" is $8 - x$,
not $x - 8$.

</div>

</div>

</section>

<!-- LESSON 3 -->

<section class="lesson"
data-category="sets"
data-search="sets number sets natural counting whole integers rational irrational real numbers
elements">

```
<button class="lesson-header">

  <div class="lesson-number">03</div>

  <div class="lesson-info">

    <div class="lesson-name">
      Sets and Number Sets
    </div>

    <div class="lesson-description">
      Types of numbers and elements
    </div>

  </div>

  <span class="arrow">v </span>

</button>
```

```
<div class="lesson-content">
```

```
<h3>What is a Set?</h3>
```

```
<p>
A set is a well-defined collection of distinct objects.
The objects in a set are called elements or members.
</p>
```

```
<div class="formula">
  A = {A, E, L, X}
</div>
```

```
<p>
The elements of set A are A, E, L, and X.
</p>
```

```
<h3>Fundamental Number Sets</h3>
```

```
<table>
```

```
<tr>
<th>Set</th>
<th>Definition</th>
<th>Example</th>
</tr>
```

```
<tr>
<td> $\mathbb{N}$  </td>
<td>Natural / Counting Numbers</td>
<td>{1, 2, 3, 4, ...}</td>
</tr>
```

```
<tr>
<td>W</td>
<td>Whole Numbers</td>
```

```
<td>{0, 1, 2, 3, ...}</td>
</tr>
```

```
<tr>
<td> $\mathbb{Z}$  </td>
<td>Integers</td>
<td>{..., -2, -1, 0, 1, 2, ...}</td>
</tr>
```

```
<tr>
<td> $\mathbb{Q}$  </td>
<td>Rational Numbers</td>
<td>1/2, 0.75, 3</td>
</tr>
```

```
<tr>
<td> </td>
<td>Irrational Numbers</td>
<td> $\sqrt{2}$ ,  $\pi$ </td>
</tr>
```

```
<tr>
<td> $\mathbb{R}$  </td>
<td>Real Numbers</td>
<td>Rational and irrational numbers</td>
</tr>
```

```
</table>
```

Examples of Sets

Set of counting numbers greater than 5 but less than 10:

```
<div class="formula">
  N = {6, 7, 8, 9}
</div>
```

Set of whole numbers greater than 0 but less than or equal to 6:

```
<div class="formula">
  W = {1, 2, 3, 4, 5, 6}
</div>
```

```
<div class="note">
<strong>Note:</strong>
Some textbooks include 0 in the natural numbers.
Your notes use  $\mathbb{N}$  = {1, 2, 3, ...}.
</div>
```

```
</div>
```

```
</section>
```

<!-- LESSON 4 -->

<section class="lesson"
data-category="sets"
data-search="set builder notation natural numbers greater less equal condition roster form">

<button class="lesson-header">

<div class="lesson-number">04</div>

<div class="lesson-info">

<div class="lesson-name">
Set-Builder Notation
</div>

<div class="lesson-description">
Writing sets using conditions
</div>

</div>

v

</button>

<div class="lesson-content">

<h3>Definition</h3>

<p>
Set-builder notation describes a set by stating
the condition that its elements must satisfy.
</p>

<div class="formula">
 $\{ x \in \mathbb{N} \mid 10 < x < 15 \}$
</div>

<p>
Read as: The set of all natural numbers x such that
x is greater than 10 but less than 15.
</p>

<h3>Example 1</h3>

<p>
Set of all natural numbers greater than 10
but less than 15.
</p>

<div class="formula">
 $\{11, 12, 13, 14\}$
</div>

Example 2

Set of all counting numbers greater than or equal to 5
but less than or equal to 10.

$$\{x \in \mathbb{N} \mid 5 \leq x \leq 10\}$$

Roster form:

{5, 6, 7, 8, 9, 10}

Symbols:

$\in$ means "is an element of."

The vertical bar $|$ means "such that."

LESSON 5

Lesson

Category: sets

Search: types of sets well defined empty finite infinite universal equal equivalent set

Lesson Header

Lesson Number: 05

Lesson Info

Lesson Name

Types of Sets

Lesson Description

Empty, finite, infinite, and other sets

Next

<div class="lesson-content">

<h3>1. Well-Defined Set</h3>

<p>

A set whose elements are clearly identified without doubt.

</p>

<div class="formula">

$$A = \{1, 2, 3, 4\}$$

</div>

<h3>2. Empty Set</h3>

<p>

A set with no elements. Also called a null set.

</p>

<div class="formula">

$$A = \varnothing \text{ or } A = \{ \}$$

</div>

<h3>3. Finite Set</h3>

<p>

A set with a limited number of elements.

</p>

<div class="formula">

$$A = \{1, 2, 3, 4, 5\}$$

</div>

<h3>4. Infinite Set</h3>

<p>

A set with endlessly continuing elements.

</p>

<div class="formula">

$$W = \{0, 1, 2, 3, \dots\}$$

</div>

<h3>5. Universal Set</h3>

<p>

The set containing all objects or elements being considered in a discussion.

</p>

<div class="formula">

$$U = \{1, 2, 3, 4, 5, 6, 7\}$$

</div>

<h3>6. Equal Sets</h3>

<p>

Two sets are equal if they contain exactly the same elements.

</p>

<div class="formula">

A = {1, 2, 3, 4}

B = {1, 2, 3, 4}

∴ A = B

</div>

<h3>7. Equivalent Sets</h3>

<p>

Two sets are equivalent if they have the same number of elements, even if their elements are different.

</p>

<div class="formula">

A = {1, 2, 3, 4}

B = {A, B, C, D}

∴ A and B are equivalent

</div>

</div>

</section>

<!-- LESSON 6 -->

<section class="lesson"

data-category="sets"

data-search="subset proper subset subset symbol equal set">

<button class="lesson-header">

<div class="lesson-number">06</div>

<div class="lesson-info">

<div class="lesson-name">

Subsets and Proper Subsets

</div>

<div class="lesson-description">

Comparing elements of sets

</div>

</div>

v

</button>

<div class="lesson-content">

Subset ($\subseteq$)

A set A is a subset of B if every element of A is also an element of B. A can be equal to B.

A = {1, 2, 3}

B = {1, 2, 3, 4, 5}

$A \subseteq B$

Proper Subset ($\subset$)

A is a proper subset of B if all elements of A are in B, but A is not equal to B.

A = {1, 2, 3, 4}

B = {1, 2, 3, 4, 5, 6, 7}

$A \subset B$

Difference:

A subset may be equal to the other set.

A proper subset must be smaller than the other set.

Example from your notes

A = {1, 2, 3, 4, 5}

B = {1, 2, 3, 4, 5, 6, 7}

$A \subset B$

!-- LESSON 7 -->

```
<section class="lesson"
data-category="sets"
data-search="venn diagram sets circles intersection union">
```

```
<button class="lesson-header">
```

```
  <div class="lesson-number">07</div>
```

```
  <div class="lesson-info">
```

```
    <div class="lesson-name">
      Venn Diagram
    </div>
```

```
    <div class="lesson-description">
      Visual representation of sets
    </div>
```

```
  </div>
```

```
  <span class="arrow">v </span>
```

```
</button>
```

```
<div class="lesson-content">
```

```
<h3>What is a Venn Diagram?</h3>
```

```
<p>
A Venn diagram is a visual representation of sets
using circles. It shows relationships between sets.
</p>
```

```
<h3>Example</h3>
```

```
<p>
A = {1, 2, 3, 4}
</p>
```

```
<p>
B = {2, 3, 5, 6}
</p>
```

```
<div class="venn-wrapper">
```

```
<svg viewBox="0 0 500 300"
xmlns="http://www.w3.org/2000/svg">
```

```
<rect x="5" y="5" width="490" height="290"
rx="15"
fill="none"
stroke="currentColor"
stroke-opacity="0.25"
stroke-width="2"/>
```

```
<text x="25" y="32"
font-size="18"
font-weight="bold"
fill="currentColor">
U
</text>
```

```
<circle cx="190" cy="155"
r="105"
fill="#60a5fa"
fill-opacity="0.35"
stroke="#3b82f6"
stroke-width="3"/>
```

```
<circle cx="310" cy="155"
r="105"
fill="#f59e0b"
fill-opacity="0.35"
stroke="#d97706"
stroke-width="3"/>
```

```
<text x="125" y="85"
font-size="24"
font-weight="bold"
fill="currentColor">
A
</text>
```

```
<text x="355" y="85"
font-size="24"
font-weight="bold"
fill="currentColor">
B
</text>
```

```
<text x="125" y="150"
font-size="21"
fill="currentColor">
1
</text>
```

```
<text x="125" y="180"
font-size="21"
fill="currentColor">
4
</text>
```

```
<text x="237" y="150"
font-size="21"
font-weight="bold"
fill="currentColor">
2
</text>
```

```
<text x="237" y="180"
font-size="21"
font-weight="bold"
```

```
fill="currentColor">
3
</text>
```

```
<text x="355" y="150"
font-size="21"
fill="currentColor">
5
</text>
```

```
<text x="355" y="180"
font-size="21"
fill="currentColor">
6
</text>
```

```
</svg>
```

```
</div>
```

<h3>Set Operations in the Diagram</h3>

```
<div class="formula">
  A ∩ B = {2, 3}
</div>
```

```
<p>
<strong>Intersection:</strong>
The elements common to both A and B.
</p>
```

```
<div class="formula">
  A ∪ B = {1, 2, 3, 4, 5, 6}
</div>
```

```
<p>
<strong>Union:</strong>
All elements in A or B or both.
</p>
```

```
<div class="note">
```

```
<strong>Remember:</strong>
```

The overlapping part is the intersection.

The two circles together represent the union.

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- LESSON 8 -->
```

```
<section class="lesson"
data-category="sets"
data-search="set operations intersection union disjoint">
```

```
<button class="lesson-header">
```

```
  <div class="lesson-number">08</div>
```

```
  <div class="lesson-info">
```

```
    <div class="lesson-name">
```

```
      Set Operations
```

```
    </div>
```

```
    <div class="lesson-description">
```

```
      Intersection, union, and disjoint sets
```

```
    </div>
```

```
  </div>
```

```
  <span class="arrow">v </span>
```

```
</button>
```

```
<div class="lesson-content">
```

```
<h3>1. Set Intersection ( $\cap$ )</h3>
```

```
<p>
```

```
The set of elements common to both A and B.
```

```
</p>
```

```
<div class="formula">
```

```
  A = {1, 2, 3, 4}<br>
```

```
  B = {2, 3, 5, 6, 7}<br><br>
```

```
  A  $\cap$  B = {2, 3}
```

```
</div>
```

```
<h3>2. Set Union ( $\cup$ )</h3>
```

```
<p>
```

```
The set of all elements belonging to A or B or both.
```

```
Repeated elements are written only once.
```

```
</p>
```

```
<div class="formula">
```

```
  A = {1, 2, 3, 4}<br>
```

```
  B = {3, 4, 5}<br><br>
```

```
  A  $\cup$  B = {1, 2, 3, 4, 5}
```

```
</div>
```

```
<h3>3. Disjoint Sets</h3>
```

```
<p>
```

```
Two sets are disjoint if they have no common elements.
```

```
</p>
```

```
<div class="formula">
  A = {1, 2, 3}<br>
  B = {4, 5, 6}<br><br>
  A ∩ B = ∅
</div>
```

```
<div class="note">
```

```
<strong>Remember:</strong>
```

Intersection means common elements.

Union means combine all elements without duplicates.

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- LESSON 9 -->
```

```
<section class="lesson"
data-category="sets"
data-search="universal set complement elements not belong">
```

```
<button class="lesson-header">
```

```
  <div class="lesson-number">09</div>
```

```
  <div class="lesson-info">
```

```
    <div class="lesson-name">
      Universal Set and Complement
    </div>
```

```
    <div class="lesson-description">
      Finding elements outside a set
    </div>
```

```
  </div>
```

```
  <span class="arrow">v </span>
```

```
</button>
```

```
<div class="lesson-content">
```

```
<h3>Universal Set (U)</h3>
```

```
<p>
The universal set contains all elements being
considered in a problem.
</p>
```

```
<div class="formula">
  U = {1, 2, 3, 4, 5, 6, 7}
</div>
```

<h3>Set Complement (A')</h3>

```
<p>
The complement of A is the set of all elements
in U that are not in A.
</p>
```

```
<div class="formula">
  U = {1, 2, 3, 4, 5, 6, 7}<br>
  A = {1, 2, 3, 4}<br><br>
  A' = {5, 6, 7}
</div>
```

<h3>Another Example</h3>

```
<p>
If U = {1, 2, 3, 4, 5} and B = {2, 3, 5},
then:
</p>
```

```
<div class="formula">
  B' = {1, 4}
</div>
```

```
<div class="note">
```

```
<strong>Tip:</strong>
```

To find the complement, start with the universal set and remove the elements that belong to A.

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- LESSON 10 -->
```

```
<section class="lesson"
data-category="reasoning"
data-search="reasoning facts information observations conclusion">
```

```
<button class="lesson-header">
```

```
  <div class="lesson-number">10</div>
```

```
  <div class="lesson-info">
```

```
    <div class="lesson-name">
      Reasoning
```

</div>

<div class="lesson-description">
Using facts to reach a conclusion
</div>

</div>

v

</button>

<div class="lesson-content">

<h3>What is Reasoning?</h3>

<p>
Reasoning is the process of using facts,
information, and observations to reach a conclusion.
</p>

<h3>Example</h3>

<div class="example">

Fact 1:
All students in the class have an ID.

Fact 2:
Juan is a student in the class.

Conclusion:
Juan has an ID.

</div>

<h3>Important Parts</h3>

Facts or information

Observations

Logical thinking

Conclusion

</div>

</section>

<!-- LESSON 11 -->

<section class="lesson"
data-category="reasoning"
data-search="deductive reasoning general specific conclusion examples">

<button class="lesson-header">

<div class="lesson-number">11</div>

<div class="lesson-info">

<div class="lesson-name">
Deductive Reasoning
</div>

<div class="lesson-description">
General to specific
</div>

</div>

v

</button>

<div class="lesson-content">

<h3>Definition</h3>

<p>
Deductive reasoning starts with a general rule
and applies it to a specific situation.
</p>

<h3>Example 1</h3>

<div class="example">

General statement:
All humans are mortal.

Specific statement:
Socrates is a human.

Conclusion:
Therefore, Socrates is mortal.

</div>

<h3>Example 2: Mathematics</h3>

<div class="example">

1. All even numbers are divisible by 2.

2. 8 is an even number.

3. Therefore, 8 is divisible by 2.

</div>

<div class="formula">

General → Specific

</div>

<div class="note">

Key idea:

Deductive reasoning applies a general rule
to a specific case.

</div>

</div>

</section>

<!-- LESSON 12 -->

<section class="lesson"

data-category="reasoning"

data-search="inductive reasoning specific general observations examples probable conclusion">

<button class="lesson-header">

<div class="lesson-number">12</div>

<div class="lesson-info">

<div class="lesson-name">

Inductive Reasoning

</div>

<div class="lesson-description">

Specific to general

</div>

</div>

```
<span class="arrow">v </span>

</button>

<div class="lesson-content">

<h3>Definition</h3>

<p>
Inductive reasoning starts with specific observations
and develops a general conclusion.
</p>

<h3>Example 1</h3>

<div class="example">

1. The sun rose in the east yesterday.

<br>

2. The sun rose in the east today.

<br>

3. Therefore, the sun will probably rise
in the east tomorrow.

</div>

<h3>Example 2: Mathematics</h3>

<div class="example">

1.  $2 + 2 = 4$ 

<br>

2.  $4 + 4 = 8$ 

<br>

3.  $6 + 6 = 12$ 

<br>

4. Therefore, adding an even number to itself
produces an even number.

</div>

<div class="formula">
Specific  $\rightarrow$  General
</div>

<div class="note">
```

Key idea:

Inductive reasoning gives a probable conclusion,
not a guaranteed one.

</div>

</div>

</section>

</div>

<div class="no-results hidden" id="noResults">

No lessons found. Try another keyword.

</div>

<!-- QUIZ -->

<section class="quiz-box">

<h2>🧠 Practice Quiz</h2>

<p class="quiz-subtitle">

Test your understanding of the lessons.

</p>

<div class="progress-track">

<div class="progress-bar" id="progressBar"></div>

</div>

<div id="quizArea">

<div class="quiz-question" id="question"></div>

<div id="options"></div>

<p class="quiz-feedback" id="feedback"></p>

<button

class="quiz-btn hidden"

id="nextBtn">

Next Question →

</button>

</div>

<div id="quizResult" class="hidden">

```
<h3> 🏆 Quiz Completed!</h3>
```

```
<p id="finalScore"></p>
```

```
<button class="quiz-btn" id="restartBtn">  
  Try Again  
</button>
```

```
</div>
```

```
</section>
```

```
</main>
```

```
<!-- FOOTER -->
```

```
<footer>
```

```
<h3>Mathematics in the Modern World</h3>
```

```
<p>  
  Developed by Franz Chezka Jacobe  
</p>
```

```
<p>  
  Student Reviewer • 2026  
</p>
```

```
</footer>
```

```
<script>
```

```
/* LESSON OPEN AND CLOSE */
```

```
const lessons = document.querySelectorAll(".lesson");
```

```
lessons.forEach(function(lesson) {
```

```
  const header = lesson.querySelector(".lesson-header");
```

```
  header.setAttribute("aria-expanded", lesson.classList.contains("open"));
```

```
  header.addEventListener("click", function() {
```

```
    const isOpen = lesson.classList.toggle("open");
```

```
    header.setAttribute("aria-expanded", isOpen);
```

```
  });
```

```
});
```

```
/* SEARCH AND FILTER */
```

```

const searchInput = document.getElementById("searchInput");

const filterButtons = document.querySelectorAll(".filter-btn");

const topicCount = document.getElementById("topicCount");

const noResults = document.getElementById("noResults");

let currentFilter = "all";

function updateLessons() {

    const searchText = searchInput.value.toLowerCase().trim();

    let visibleCount = 0;

    lessons.forEach(function(lesson) {

        const category = lesson.dataset.category;

        const searchData = (
            lesson.dataset.search + " " +
            lesson.textContent
        ).toLowerCase();

        const matchesFilter =
            currentFilter === "all" ||
            category === currentFilter;

        const matchesSearch =
            searchData.includes(searchText);

        const show = matchesFilter && matchesSearch;

        lesson.classList.toggle("hidden", !show);

        if (show) {
            visibleCount++;
        }

    });

    topicCount.textContent =
        visibleCount + " Topic" +
        (visibleCount === 1 ? "" : "s");

    noResults.classList.toggle(
        "hidden",
        visibleCount !== 0
    );

}

searchInput.addEventListener("input", updateLessons);

```

```

filterButtons.forEach(function(button) {

    button.addEventListener("click", function() {

        currentFilter = button.dataset.filter;

        filterButtons.forEach(function(btn) {

            btn.classList.remove("active");

        });

        button.classList.add("active");

        updateLessons();

    });

});

/* DARK MODE */

const themeBtn = document.getElementById("themeBtn");

themeBtn.addEventListener("click", function() {

    document.body.classList.toggle("dark");

    if (document.body.classList.contains("dark")) {

        themeBtn.textContent = "☀ Light Mode";

    } else {

        themeBtn.textContent = "☾ Dark Mode";

    }

});

/* QUIZ */

const questions = [

    {

        question:

        "What is the intersection of A = {1, 2, 3} and B = {2, 3, 4}?",

        options: [

            "{1, 2, 3, 4}",

            "{2, 3}",

            "{1, 4}",

            "∅ "

        ],

    },


```

answer: 1,

explanation:

"Intersection contains the common elements: 2 and 3."

},

{

question:

"What is the expression for 'the product of 10 and a number'?",

options: [

" $x + 10$ ",

" $x - 10$ ",

" $10x$ ",

" $x / 10$ "

],

answer: 2,

explanation:

"Product means multiplication, so $10 \times x = 10x$."

},

{

question:

"Which set contains all natural numbers greater than 5 but less than 10?",

options: [

"{5, 6, 7, 8, 9, 10}",

"{6, 7, 8, 9}",

"{1, 2, 3, 4}",

"{5, 10}"

],

answer: 1,

explanation:

"Greater than 5 and less than 10 gives 6, 7, 8, and 9."

},

{

question:

"If $U = \{1, 2, 3, 4, 5\}$ and $A = \{1, 2\}$, what is A' ?",

options: [

"{1, 2}",

"{3, 4, 5}",

" $\emptyset$ ",

"{1, 2, 3, 4, 5}"

],

answer: 1,

explanation:

"The complement contains elements in U that are not in A ."

},

```
{  
  question:  
    "Which symbol means 'less than or equal to'?",
```

```
  options: [
```

```
    "≥",
```

```
    ">",
```

```
    "≤",
```

```
    "≠"
```

```
  ],
```

```
  answer: 2,
```

```
  explanation:
```

```
    "≤ means less than or equal to."
```

```
},
```

```
{
```

```
  question:
```

```
    "Deductive reasoning moves from:",
```

```
  options: [
```

```
    "Specific to general",
```

```
    "General to specific",
```

```
    "Unknown to unknown",
```

```
    "Opinion to fact"
```

```
  ],
```

```
  answer: 1,
```

```
  explanation:
```

```
    "Deductive reasoning moves from a general rule to a specific conclusion."
```

```
},
```

```
{
```

```
  question:
```

```
    "Inductive reasoning usually gives a conclusion that is:",
```

```
  options: [
```

```
    "Always guaranteed",
```

```
    "Impossible",
```

```
    "Probable",
```

```
    "Always false"
```

```
  ],
```

```
  answer: 2,
```

```
  explanation:
```

```
    "Inductive reasoning uses observations to form a probable conclusion."
```

```
},
```

```
{
```

```
  question:
```

```
    "What is the union of  $A = \{1, 2\}$  and  $B = \{2, 3\}$ ?",
```

```
  options: [
```

```

        "{2}",
        "{1, 2, 3}",
        "{1, 3}",
        "∅ "
    ],

    answer: 1,

    explanation:
    "Union combines all elements without repeating 2."
}

];

```

```
let currentQuestion = 0;
```

```
let score = 0;
```

```
let answered = false;
```

```
const questionElement =
    document.getElementById("question");
```

```
const optionsElement =
    document.getElementById("options");
```

```
const feedbackElement =
    document.getElementById("feedback");
```

```
const nextBtn =
    document.getElementById("nextBtn");
```

```
const progressBar =
    document.getElementById("progressBar");
```

```
const quizArea =
    document.getElementById("quizArea");
```

```
const quizResult =
    document.getElementById("quizResult");
```

```
const finalScore =
    document.getElementById("finalScore");
```

```
const restartBtn =
    document.getElementById("restartBtn");
```

```
function showQuestion() {
```

```
    const item = questions[currentQuestion];
```

```
    answered = false;
```

```
    questionElement.textContent = item.question;
```

```

optionsElement.innerHTML = "";

feedbackElement.textContent = "";

nextBtn.classList.add("hidden");

progressBar.style.width =
  (currentQuestion / questions.length * 100) + "%";

item.options.forEach(function(option, index) {

  const button = document.createElement("button");

  button.className = "option";

  button.textContent =
    String.fromCharCode(65 + index) +
    ". " + option;

  button.addEventListener("click", function() {

    checkAnswer(index);

  });

  optionsElement.appendChild(button);

});

}

function checkAnswer(selectedIndex) {

  if (answered) return;

  answered = true;

  const item = questions[currentQuestion];

  const optionButtons =
    optionsElement.querySelectorAll(".option");

  if (selectedIndex === item.answer) {

    score++;

  }

  optionButtons.forEach(function(button, index) {

    button.disabled = true;

```

```

    if (index === item.answer) {

        button.classList.add("correct");

    }

    if (
        index === selectedIndex &&
        selectedIndex !== item.answer
    ) {

        button.classList.add("wrong");

    }

});

if (selectedIndex === item.answer) {

    feedbackElement.textContent =
        "✓ Correct! " + item.explanation;

} else {

    feedbackElement.textContent =
        "✗ Not quite. " + item.explanation;

}

progressBar.style.width =
    ((currentQuestion + 1) / questions.length * 100) + "%";

nextBtn.classList.remove("hidden");

}

nextBtn.addEventListener("click", function() {

    currentQuestion++;

    if (currentQuestion >= questions.length) {

        quizArea.classList.add("hidden");

        quizResult.classList.remove("hidden");

        finalScore.textContent =
            "You scored " + score +
            " out of " + questions.length + "!";

    } else {

        showQuestion();
    }
});

```

```
    }  
  });  
  
  restartBtn.addEventListener("click", function() {  
    currentQuestion = 0;  
    score = 0;  
    quizResult.classList.add("hidden");  
    quizArea.classList.remove("hidden");  
    showQuestion();  
  });  
  
  /* INITIALIZE */  
  updateLessons();  
  showQuestion();  
  
</script>  
  
</body>  
</html>
```